Cedric Legara — Gameplay Programmer & Unity Engineer

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Summary

Computer Science graduate specializing in gameplay engineering, system architecture, and mobile optimization. Experienced in taking ownership of complex Unity codebases, refactoring scalable gameplay systems, and shipping published interactive applications across PC, mobile, and web platforms. Brings hybrid experience in game development and full-stack software engineering, with strong interests in technical architecture, rapid prototyping, and player-focused system design.

Projects

Metrocycle 2.0 (Road-Safety Educational Game) Aug 2024 — Jul 2025
Team of 3 | Published (WCTP 2025 Proceedings @ Atlantis Press; itch.io), Thesis, Unity 3D, C#, Android

- Took technical ownership of Metrocycle, a legacy web-based Unity codebase, extending its core systems to deliver a production-ready Android educational application.
- Translated research-backed educational goals into scalable systems, prioritizing clean architecture for future development teams.
- Architected a centralized localization framework by replacing hardcoded text with a scalable, data-driven key-reference architecture parsed directly from CSV files.
- Engineered highly extensible gameplay frameworks using ScriptableObjects, delivering a modular mix-and-match weather system and a plug-and-play architecture for adding 2D minigames.
- Managed the Android migration pipeline, debugging and resolving critical cross-platform compatibility issues within a third-party AI traffic package to restore core gameplay loops on mobile.
- Co-managed GitHub version control workflows, rigorous technical documentation, and agile team practices to deliver an application validated by real riders and domain experts.
- Presented at the Workshop on Computation: Theory and Practice (WCTP) 2025 in Hokkaido, Japan, and published the technical architecture and findings in the conference proceedings (Atlantis Press) following rigorous academic peer review.

Odd-One-Out (Endless Runner Word-Puzzle Game) Apr 2025 — Jun 2025
Solo Developer | Published (Google Play Store, itch.io), Unity 3D, C#, Mobile Monetization

- Designed and engineered fast-paced endless runner mechanics and core gameplay loops with scalable, data-driven level design.
- Integrated third-party monetization SDKs (IronSource, Unity Ads), implementing rewarded, banner, and interstitial ad flows.
- Conducted closed beta testing with 10+ external testers, iterating on gameplay balance, UX clarity, and progression pacing.
- Managed the full mobile release pipeline from prototyping and playtesting to Google Play Store deployment.

Life as a Marites (Story-Driven Adventure Game) Jan 2025 — Jun 2025
Team of 2 | Published (itch.io), Unity 3D, C#, Data-Driven UI

- Designed and implemented a data-driven dialogue system to support complex branching narratives, dynamic state tracking, and player consequence logic.
- Developed responsive UI architectures handling dynamic text parsing and event-driven player feedback systems.
- Coordinated with a designer to structure branching narratives and player decision consequences.
- Maintained a detailed Game Design Document (GDD) covering gameplay systems, narrative structure, timelines, and production.

Heavenly Hatdogs (Physics Platformer) Apr 2022 — May 2022
Solo Developer | Published (itch.io), Demoed at ConQuest 2023, Unity 3D, C#, Physics Mechanics

- Programmed a custom physics-based kinematic character controller utilizing raycasting and vector mathematics for precise slope handling, grounded states, and custom gravity and time manipulation.
- Iterated core gameplay mechanics from a rapid game jam prototype into a polished, convention-ready vertical slice optimized for live player usability testing.
- Presented the game at ConQuest 2023, gathering direct player usability feedback for later gameplay improvements.

Skills

- **Game Development:** Unity (2D/3D), C#, Gameplay Systems, Custom Physics Controllers, Technical Prototyping, Mobile Optimization, Data-Driven Architecture
- **Web & Backend (Full-Stack):** TypeScript, JavaScript, Python, SvelteKit, React Native, PostgreSQL, Supabase, REST APIs
- **Tools & Pipelines:** Git, GitHub, Unity Version Control (PlasticSCM), Google Play Console, Android SDK, Debugging
- **Monetization & Systems:** IronSource SDK, Unity Ads, Localization Frameworks, UI/UX Implementation, Agile Methodology

Education

Bachelor of Science in Computer Science Aug 2022 — Jul 2026
University of the Philippines Diliman

- GWA: 1.68 (Cum Laude equivalent) | Department of Science & Technology (DOST) Merit Scholar